

ALBUM OPERATIONS CRM

Design + Album Printing & Album Printing Only

Fully Traceable Software Requirements Specification

PROJECT	Album Design & Printing Operations CRM / ERP
DOCUMENT	All-in-One Role-Based Software Requirements Specification
VERSION	1.0
DATE	09 October 2026
ROLES	Admin, Reception, Colour Grading, Designing, Printing, QC, Accounts
AUDIENCE	Product, UI/UX, Engineering, QA, UAT & Operations

Scope principle: This document defines the complete operational lifecycle for an album company from customer/order intake through colour grading, album designing, client review and corrections, printing, quality control, accounts and final delivery. Every state-changing activity is traceable through role, timestamp, status, files, approvals and audit history.

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1. Document Purpose, Scope & Role Model

1.1 Purpose

This specification defines field-level, action-level, workflow-level and control requirements for an album design and printing company. The CRM must support two commercial service models: Album Printing Only and Design + Album Printing. All requirements are written so they can be traced into UI/UX, APIs, development stories, QA test cases and UAT.

1.2 Role Boundaries

Role	Primary Scope	Explicit Boundary
Admin	Full operational control, approvals, assignment, configuration, override, reports and final closure.	May view and control all modules subject to security policy.
Reception	Customer registration, order creation, requirement capture, file intake, basic payment receipt and status enquiry.	Cannot approve designs/QC, alter protected finance records or override production.
Colour Grading	Image grading queue, processing, upload, submit for approval, revisions.	Cannot approve own work or modify commercial/order pricing.
Designing	Album design, revisions, proof preparation and submission.	Cannot self-approve final design or release to printing.
Printing	Approved print job execution, material/production stage updates, packaging and transfer to QC.	Cannot modify approved artwork or customer finance.
QC	Inspection, evidence, pass/fail/rework decisions and transfer to delivery.	Cannot modify final design or commercial terms.
Accounts	Quotation/invoice/payment/dues/refund/expense controls and financial reporting.	Cannot alter creative/production files or QC decisions.

1.3 Shared Operating Principles

Requirement ID	Functional Requirement / Business Rule
ALB-FR-0001	Every user shall operate within assigned role, department, branch and permission scope.
ALB-FR-0002	Every state-changing action shall capture user ID, role, timestamp, source screen, previous state, new state and reason when applicable.
ALB-FR-0003	Sensitive customer phone, email, tax and payment data shall be masked when the user lacks explicit permission.
ALB-FR-0004	Lists shall support search, filter, sort, pagination, saved views and export subject to permission.
ALB-FR-0005	All approvals, rejections, corrections, rework and overrides shall be traceable and immutable after finalization except through controlled amendment.
ALB-FR-0006	SLA timers shall use server time and configured business calendars; pause/resume reasons must be traceable.
ALB-FR-0007	Attachments shall be type/size validated, access controlled and versioned where applicable.
ALB-FR-0008	No department shall silently bypass another department's required approval or handoff.

2. Authentication, Login, MFA & Session Security

2.1 Login Fields

Requirement ID	Field / Column	Type	M/O	Validation / Allowed Values	Functional Behaviour
ALB-FR-0009	Work Email / Mobile	Text	M	Valid registered identifier	Identifies authenticated employee.
ALB-FR-0010	Password	Password	M	Configured password policy	Authenticates credential.
ALB-FR-0011	Remember Device	Checkbox	O	Trusted device policy	Reduces MFA prompts where allowed.
ALB-FR-0012	CAPTCHA	Challenge	C	Risk-based	Mitigates automated abuse.

2.2 MFA & Session Actions

Requirement ID	Action / Control	Permission / Availability	Functional Behaviour
ALB-FR-0013	Login	Active user	Validate credential, status, role, MFA and session policy before access.
ALB-FR-0014	Forgot Password	Registered user	Issue single-use, time-bound recovery flow.
ALB-FR-0015	Logout	Authenticated user	Invalidate current session.
ALB-FR-0016	Logout All Sessions	Permitted user	Invalidate other sessions after re-authentication.
ALB-FR-0017	View Active Sessions	Authenticated user	Show device/browser, login time and last activity.
ALB-FR-0018	Step-up Authentication	High-risk actions	Require OTP/re-authentication before sensitive export, override or permission change.

3. Master Data & Configuration

3.1 Configurable Masters

Requirement ID	Master	Examples	Functional Behaviour
ALB-FR-0019	Album Type	Photo Book, Acrylic Album, Premium Album, Coffee Table	Used in order requirement capture and pricing.
ALB-FR-0020	Album Size	8x12, 12x30, 14x40, custom	Controls design template and print specification.
ALB-FR-0021	Paper Type	Silk, Metallic, Matte, Glossy, Velvet, Fine Art	Printing material selection.
ALB-FR-0022	Paper GSM	Configured numeric values	Validates print stock.
ALB-FR-0023	Cover Type	Acrylic, Leatherette, Photo Wrap, Fabric, Custom	Production requirement.
ALB-FR-0024	Lamination	Matte, Gloss, Texture, None	Printing/finishing requirement.
ALB-FR-0025	Binding Type	Lay-flat, Flush mount, Standard, Custom	Manufacturing routing.
ALB-FR-0026	Box Type	Standard, Premium, Wooden, Acrylic, Custom	Packaging specification.
ALB-FR-0027	Event Type	Wedding, Reception, Engagement, Baby, Corporate, Other	Order categorization.
ALB-FR-0028	Priority	Normal, High, Urgent, VIP	Queue and SLA sequencing.
ALB-FR-0029	QC Defect	Colour, scratch, alignment, binding, print, damage, wrong spec	Standardized QC reason taxonomy.
ALB-FR-0030	Delivery Mode	Pickup, company delivery, courier, third-party logistics	Final dispatch routing.

3.2 Pricing Rules

Requirement ID	Rule / Field	Type	Functional Behaviour
ALB-FR-0031	Base Album Price	Currency	Default price by album type and size.
ALB-FR-0032	Per Sheet / Page Price	Currency	Adds variable charge based on selected sheet count.
ALB-FR-0033	Design Charge	Currency / formula	Applies when order type includes design.
ALB-FR-0034	Colour Grading Charge	Currency / formula	Configurable fixed/per-image/per-order charge.
ALB-FR-0035	Cover / Box Add-on	Currency	Applies selected premium materials.
ALB-FR-0036	Discount Rule	Amount/%	Requires permission and optional approval threshold.
ALB-FR-0037	GST / Tax	Percentage	Applies configured tax rules and invoice breakup.

4. Reception Dashboard, Customer & Order Intake

4.1 Reception Dashboard

Requirement ID	Widget / Queue	Type	Functional Behaviour
ALB-FR-0038	New Orders Today	KPI	Count orders created today.
ALB-FR-0039	Files Pending	KPI	Orders where required source files are not complete.
ALB-FR-0040	Payment Pending	KPI	Orders with outstanding amount visible to Reception if permitted.
ALB-FR-0041	Ready for Pickup	KPI	QC-passed orders awaiting customer pickup.
ALB-FR-0042	Due Today	KPI	Orders with promised delivery date today.
ALB-FR-0043	Recent Orders	Table	Quick access to customer/order status.

4.2 Customer Creation Fields

Requirement ID	Field / Column	Type	M/O	Validation / Allowed Values	Functional Behaviour
ALB-FR-0044	Customer ID	Auto ID	M	Unique	Primary customer identifier.
ALB-FR-0045	Customer / Studio Name	Text	M	2-150 chars	Customer identity.
ALB-FR-0046	Contact Person	Text	O	Safe text	Primary person for studio account.
ALB-FR-0047	Mobile Number	Phone	M	Valid phone	Primary contact.
ALB-FR-0048	WhatsApp Number	Phone	O	Valid phone	Used for proof/notification if consented.
ALB-FR-0049	Email	Email	O	Valid email	Used for proofing and documents.
ALB-FR-0050	Address	Textarea	O	Safe text	Billing/delivery address.
ALB-FR-0051	City / State / PIN	Structured	O	Valid values	Supports invoicing and logistics.
ALB-FR-0052	GSTIN / Tax ID	Text	O	Format validated	Used for tax invoice.
ALB-FR-0053	Customer Notes	Textarea	O	Internal notes	Operational context.

4.3 Order Creation Fields

Requirement ID	Field / Column	Type	M/O	Validation / Allowed Values	Functional Behaviour
ALB-FR-0054	Order ID	Auto ID	M	Unique	Primary lifecycle identifier.
ALB-FR-0055	Customer	Search select	M	Existing/new customer	Links order to customer.
ALB-FR-0056	Order Type	Dropdown	M	Album Printing Only /	Determines workflow route.

Requirement ID	Field / Column	Type	M/O	Validation / Allowed Values	Functional Behaviour
				Design + Album Printing	
ALB-FR-0057	Order Date	Date	M	Default today	Commercial intake date.
ALB-FR-0058	Expected Delivery Date	Date	M	Future/override reason	Controls SLA and priority.
ALB-FR-0059	Event Name	Text	O	Safe text	Human-readable event label.
ALB-FR-0060	Event Type	Dropdown	O	Configured master	Reporting and categorization.
ALB-FR-0061	Bride Name	Text	O	Safe text	Wedding context.
ALB-FR-0062	Groom Name	Text	O	Safe text	Wedding context.
ALB-FR-0063	Event Date	Date	O	Valid date	Event reference.
ALB-FR-0064	Priority	Dropdown	M	Normal/High/Urgent/VIP	Queue sequencing.
ALB-FR-0065	Special Instructions	Textarea	O	Safe text	Customer production notes.

5. Order Types & Workflow Routing

5.1 Design + Album Printing

Default route: Reception -> Colour Grading -> Admin Approval -> Designing -> Admin Design Review -> Client Review / Corrections -> Final Client Approval -> Printing -> QC -> Delivery -> Order Closure.

Requirement ID	Step	Workflow Behaviour
ALB-FR-0066	Step 1	Reception captures customer, order and printing requirement; source files are received.
ALB-FR-0067	Step 2	Order enters Colour Grading queue and is assigned to an eligible staff member.
ALB-FR-0068	Step 3	Colour grading work is submitted to Admin for approval; rejection creates revision task.
ALB-FR-0069	Step 4	Approved grading moves to Designing; designer creates album draft.
ALB-FR-0070	Step 5	Designer submits draft to Admin; Admin approves or sends correction.
ALB-FR-0071	Step 6	Admin-approved proof is shared with client by secure link/WhatsApp/email.
ALB-FR-0072	Step 7	Client may approve or request page-level corrections; revisions loop until final approval.
ALB-FR-0073	Step 8	Final approval locks print version and releases Printing job.
ALB-FR-0074	Step 9	Printing updates production stages and transfers completed package to QC.
ALB-FR-0075	Step 10	QC passes or creates rework; passed order becomes Ready for Delivery.
ALB-FR-0076	Step 11	Delivery is completed and order may be closed after financial and operational checks.

5.2 Album Printing Only

Default route: Reception -> File Verification -> Printing -> QC -> Delivery -> Order Closure. Colour Grading and Designing are skipped unless Admin explicitly inserts them as exception tasks.

Requirement ID	Business Rule
ALB-FR-0077	Printing-only orders shall require a final print-ready file before release to Printing.
ALB-FR-0078	If the supplied file is incomplete, low-resolution or not compliant with specification, the order shall be placed on hold and Reception/Admin notified.
ALB-FR-0079	Skipped workflow stages shall remain visible as Not Required for reporting and audit.
ALB-FR-0080	Admin may convert an order from Printing Only to Design + Printing with a reason and revised commercial approval.

6. Printing Requirement Capture

6.1 Album Specification Fields

Requirement ID	Field / Column	Type	M/O	Validation / Allowed Values	Functional Behaviour
ALB-FR-0081	Album Type	Dropdown	M	Configured master	Defines product category.
ALB-FR-0082	Album Size	Dropdown/custom	M	Configured/custom	Controls design canvas and production.
ALB-FR-0083	Orientation	Dropdown	M	Landscape/Portrait/Square	Design and print layout.
ALB-FR-0084	Pages / Sheets	Number	M	Positive integer	Production quantity basis.
ALB-FR-0085	Copies	Number	M	>=1	Number of physical albums.
ALB-FR-0086	Paper Type	Dropdown	M	Configured master	Print stock.
ALB-FR-0087	Paper GSM	Dropdown/number	C	Mapped to paper	Material validation.
ALB-FR-0088	Printing Type	Dropdown	M	Configured	Digital/photographic/other process.
ALB-FR-0089	Cover Type	Dropdown	M	Configured master	Physical cover requirement.
ALB-FR-0090	Cover Material	Dropdown	O	Configured	Material selection.
ALB-FR-0091	Cover Colour	Colour/text	O	Configured/custom	Cover appearance.
ALB-FR-0092	Lamination	Dropdown	O	Configured	Finishing requirement.
ALB-FR-0093	Binding	Dropdown	M	Configured	Manufacturing method.
ALB-FR-0094	Sheet Thickness	Dropdown/number	O	Configured	Mounting/board requirement.
ALB-FR-0095	Box Required	Yes/No	M	Boolean	Determines box sub-workflow.
ALB-FR-0096	Box Type	Dropdown	C	Required if box=yes	Packaging specification.
ALB-FR-0097	UV Printing	Yes/No	O	Boolean	Special finish.
ALB-FR-0098	Foiling	Yes/No	O	Boolean	Special finish.
ALB-FR-0099	Embossing	Yes/No	O	Boolean	Special finish.
ALB-FR-0100	Acrylic Cover	Yes/No	O	Boolean	Special cover flag.
ALB-FR-0101	Name Printing	Text	O	Safe text	Customer-visible personalized text.
ALB-FR-0102	Logo Printing	Attachment / select	O	Image rules	Branding asset for cover/box.
ALB-FR-0103	Special Instructions	Textarea	O	Safe text	Production notes.

7. File, Asset & Version Management

7.1 File Register

Requirement ID	Field / Column	Type	M/O	Validation / Allowed Values	Functional Behaviour
ALB-FR-0104	File ID	Auto ID	M	Unique	Traceable file record.
ALB-FR-0105	Order ID	Link	M	Valid order	Associates file to job.
ALB-FR-0106	File Category	Dropdown	M	Source Photos / Graded / Design Draft / Final Print / Cover / QC Evidence / Invoice / Other	Controls access and lifecycle.
ALB-FR-0107	File Name	Read-only	M	Safe filename	Displays uploaded file.
ALB-FR-0108	Format	Read-only	M	Allowed extension	Validates compatibility.
ALB-FR-0109	File Size	Read-only	M	Configured limits	Supports storage controls.
ALB-FR-0110	Version	Auto number	M	Incrementing	Maintains revision history.
ALB-FR-0111	Uploaded By	Read-only	M	Authenticated user	Audit ownership.
ALB-FR-0112	Uploaded At	Datetime	M	Server time	Audit timestamp.
ALB-FR-0113	Approval State	Status	O	Draft/Submitted/Approved/Rejected/Locked	Controls workflow usage.
ALB-FR-0114	Cloud Link / Storage Path	Secure link	O	Permission controlled	Supports external storage integration.
ALB-FR-0115	Checksum	System	O	Generated	Detects accidental duplicate/changed file.

7.2 File Actions & Rules

Requirement ID	Action / Control	Permission / Availability	Functional Behaviour
ALB-FR-0116	Upload Files	Role-based	Validate extension/size and link to order.
ALB-FR-0117	Download	View permission	Download only permitted file categories.
ALB-FR-0118	Preview	View permission	Browser preview where format supports.
ALB-FR-0119	Replace / New Version	Create version permission	Never overwrite finalized version silently.
ALB-FR-0120	Lock Final Print File	Admin after final client approval	Marks exact file released to Printing.
ALB-FR-0121	Archive	Admin	Remove from active lists while retaining audit.
ALB-FR-0122	Storage Usage View	Admin	Shows order/customer/global usage and quota.

8. Colour Grading Department

8.1 Dashboard & Queue

Requirement ID	Widget / Filter	Type	Functional Behaviour
ALB-FR-0123	New Jobs	KPI	Unstarted assigned grading jobs.
ALB-FR-0124	In Progress	KPI	Jobs currently being worked.
ALB-FR-0125	Pending Admin Approval	KPI	Submitted grading awaiting decision.
ALB-FR-0126	Revision Required	KPI	Admin-rejected jobs needing changes.
ALB-FR-0127	Completed	KPI	Approved grading jobs.
ALB-FR-0128	Due Today / Overdue	KPI	SLA-based urgency counts.
ALB-FR-0129	Assigned To	Filter	Filters queue by eligible colour grader.
ALB-FR-0130	Priority	Filter	Normal/High/Urgent/VIP.
ALB-FR-0131	Due Date	Filter	Date-range view of work.

8.2 Job Detail Fields

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0132	Order ID	Read-only link	M	Opens permitted order context.
ALB-FR-0133	Customer / Event	Read-only	M	Shows customer and event context.
ALB-FR-0134	Source Image Count	Number	M	Shows workload quantity.
ALB-FR-0135	Source Files	Attachment list	M	Downloads original selection.
ALB-FR-0136	Assigned Grader	User	M	Current owner.
ALB-FR-0137	Due At	Datetime	M	SLA deadline.
ALB-FR-0138	Grading Instructions	Textarea	O	Admin/Reception notes.
ALB-FR-0139	Output Files	Upload	M	Stores graded output.
ALB-FR-0140	Internal Note	Textarea	O	Department note.
ALB-FR-0141	Current Status	Status	M	Awaiting/In Progress/Submitted/Revision/Approved.

8.3 Actions

Requirement ID	Action	Functional Behaviour
ALB-FR-0142	Start Grading	Records start timestamp and sets In Progress.
ALB-FR-0143	Save Draft	Stores progress without submission.
ALB-FR-0144	Upload Graded Files	Adds output version(s) to order.

Requirement ID	Action	Functional Behaviour
ALB-FR-0145	Submit to Admin	Validates required files and sends approval task.
ALB-FR-0146	Acknowledge Revision	Accepts admin correction request and reopens work.
ALB-FR-0147	Resubmit	Creates new submitted version while retaining previous history.

9. Album Designing Department

9.1 Designer Dashboard

Requirement ID	Widget / Queue	Type	Functional Behaviour
ALB-FR-0148	Awaiting Design	KPI	Jobs ready after approved grading.
ALB-FR-0149	Designing	KPI	Jobs currently being designed.
ALB-FR-0150	Pending Admin Review	KPI	Drafts submitted to admin.
ALB-FR-0151	Client Correction	KPI	Jobs returned with client feedback.
ALB-FR-0152	Admin Correction	KPI	Jobs returned by admin.
ALB-FR-0153	Final Approved	KPI	Locked designs released to printing.
ALB-FR-0154	Due Today / Overdue	KPI	Highlights deadline risk.

9.2 Design Workspace

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0155	Order / Customer	Read-only	M	Shows order context.
ALB-FR-0156	Album Size / Sheets	Read-only	M	Defines layout target.
ALB-FR-0157	Printing Specification	Read-only panel	M	Shows paper, binding, cover and finishing.
ALB-FR-0158	Colour-Graded Files	Attachment list	M	Input asset source.
ALB-FR-0159	Reference Album / Style	Attachment/link	O	Visual style reference.
ALB-FR-0160	Design Instructions	Textarea	O	Admin/customer design notes.
ALB-FR-0161	Current Revision	Read-only	M	Shows active revision number.
ALB-FR-0162	Draft Upload	Upload	M	Stores proof PDF/JPG or project output.
ALB-FR-0163	Page Count	Number	M	Validates against order specification.
ALB-FR-0164	Designer Note	Textarea	O	Internal submission note.

9.3 Designer Actions

Requirement ID	Action	Functional Behaviour
ALB-FR-0165	Start Designing	Records start time and owner.
ALB-FR-0166	Save Draft	Saves working revision.
ALB-FR-0167	Submit to Admin	Creates admin review task.
ALB-FR-0168	Open Correction List	Shows page-wise feedback and unresolved items.
ALB-FR-0169	Mark Correction Resolved	Marks individual correction resolved with note.

Requirement ID	Action	Functional Behaviour
ALB-FR-0170	Resubmit Revision	Creates next revision and sends to admin.
ALB-FR-0171	View Version History	Shows previous drafts, approvals and client feedback.

10. Admin Review & Approval

10.1 Colour Grading Approval

Requirement ID	Action / Field	Permission	Functional Behaviour
ALB-FR-0172	Preview Graded Output	Admin	Review submitted graded files.
ALB-FR-0173	Approve Grading	Admin	Moves order to Designing queue.
ALB-FR-0174	Request Correction	Admin	Requires structured reason, note and optional evidence; returns to grader.
ALB-FR-0175	Reassign	Admin	Moves work to another eligible colour grader with audit.
ALB-FR-0176	Override Due Date	Admin	Requires reason and updates SLA.

10.2 Design Approval

Requirement ID	Action / Field	Permission	Functional Behaviour
ALB-FR-0177	Preview Album Proof	Admin	Review proof before client sees it.
ALB-FR-0178	Approve for Client Review	Admin	Publishes selected revision to secure client link.
ALB-FR-0179	Return to Designer	Admin	Creates page-level/general correction.
ALB-FR-0180	Attach Markup	Admin	Allows annotated screenshot/PDF evidence.
ALB-FR-0181	Set Client Response Due	Admin	Sets follow-up reminder/SLA.
ALB-FR-0182	Approve Directly for Printing	Elevated Admin	Allowed only when documented client approval exists or policy permits; reason required.

11. Client Proofing, Approval & Correction Cycle

11.1 Proof Distribution

Requirement ID	Field / Action	Type	Functional Behaviour
ALB-FR-0183	Secure Proof Link	Generated URL	Tokenized, expiring, permission-controlled proof access.
ALB-FR-0184	Send via WhatsApp	Action	Sends approved template with proof link and logs delivery attempt.
ALB-FR-0185	Send via Email	Action	Sends branded email with link and logs status.
ALB-FR-0186	Copy Link	Action	Allows manual sharing and records generation.
ALB-FR-0187	Expiry Date	Datetime	Controls proof availability.
ALB-FR-0188	Client Authentication	OTP / token	Optional policy to verify reviewer.

11.2 Client Review Fields & Actions

Requirement ID	Field / Action	Type	Functional Behaviour
ALB-FR-0189	Page Thumbnail / Preview	Gallery	Displays proof pages in sequence.
ALB-FR-0190	Approve Page	Action	Marks page accepted for current revision.
ALB-FR-0191	Request Correction	Action	Creates page-specific correction with comment.
ALB-FR-0192	General Comment	Textarea	Captures overall feedback.
ALB-FR-0193	Upload Reference	Attachment	Client may attach reference screenshot/image if permitted.
ALB-FR-0194	Approve Entire Album	Action	Requires confirmation and records final approval.
ALB-FR-0195	Approval Declaration	Checkbox	Customer confirms reviewed content and printing authorization.
ALB-FR-0196	Approval Metadata	System	Captures time, version, IP/device metadata per policy.

11.3 Correction Management

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0197	Correction ID	Auto ID	M	Unique correction record.
ALB-FR-0198	Revision	Number	M	Links correction to proof version.
ALB-FR-0199	Page Number	Number	C	Required for page-specific issue.
ALB-FR-0200	Client Comment	Textarea	M	Original customer feedback retained.
ALB-FR-0201	Admin Clarification	Textarea	O	Internal interpretation if needed.

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0202	Reference Attachment	Attachment	O	Supporting image/markup.
ALB-FR-0203	Assigned Designer	User	M	Correction owner.
ALB-FR-0204	Status	Status	M	Received/In Progress/Resolved/Admin Review/Sent to Client.
ALB-FR-0205	Resolved Note	Textarea	C	Required when closing correction.
ALB-FR-0206	Resolved At	Datetime	C	Timestamp on completion.

11.4 Final Approval & Print Lock

Requirement ID	Business Rule
ALB-FR-0207	Final client approval shall lock the approved design revision as the only default print source.
ALB-FR-0208	Any post-approval design change shall require Admin reopen action, reason, new revision and renewed client approval unless policy explicitly allows internal non-content correction.
ALB-FR-0209	Printing shall never see an unapproved design as Ready to Print.
ALB-FR-0210	The system shall retain the exact version approved by the client and the exact version released to printing.

12. Printing Department & Production Tracking

12.1 Printing Dashboard

Requirement ID	Widget / Queue	Type	Functional Behaviour
ALB-FR-0211	Waiting for Printing	KPI	Approved jobs not started.
ALB-FR-0212	Print File Preparation	KPI	Jobs being prepared for printer.
ALB-FR-0213	Printing Started	KPI	Active print production.
ALB-FR-0214	Finishing / Binding	KPI	Jobs in downstream manufacturing.
ALB-FR-0215	Packaging	KPI	Jobs in packaging.
ALB-FR-0216	Sent to QC	KPI	Jobs handed off to QC.
ALB-FR-0217	Delayed Jobs	KPI	Past production due date.
ALB-FR-0218	Today Target	KPI	Planned production count for shift/day.

12.2 Printing Job Detail

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0219	Order ID	Link	M	Primary production job identifier.
ALB-FR-0220	Customer	Read-only	M	Client context.
ALB-FR-0221	Final Approved Design	Locked file	M	Only approved print version downloadable.
ALB-FR-0222	Cover File	Locked file	C	Required where separate cover artwork exists.
ALB-FR-0223	Album Size	Read-only	M	Print dimension.
ALB-FR-0224	Sheets / Pages	Read-only	M	Quantity specification.
ALB-FR-0225	Copies	Read-only	M	Production quantity.
ALB-FR-0226	Paper Type / GSM	Read-only	M	Material requirement.
ALB-FR-0227	Printing Type	Read-only	M	Process requirement.
ALB-FR-0228	Binding / Lamination	Read-only	M	Finishing specification.
ALB-FR-0229	Cover / Box Requirement	Read-only	O	Packaging and finish details.
ALB-FR-0230	Special Instructions	Read-only	O	Production notes.
ALB-FR-0231	Assigned Operator	User	M	Job owner.
ALB-FR-0232	Production Due At	Datetime	M	Printing SLA deadline.

12.3 Production Stages

Requirement ID	Stage	Functional Behaviour
ALB-FR-0233	Waiting for Printing	Record stage start/completion timestamps, operator,

Requirement ID	Stage	Functional Behaviour
		remarks and exceptions for waiting for printing.
ALB-FR-0234	Print File Prepared	Record stage start/completion timestamps, operator, remarks and exceptions for print file prepared.
ALB-FR-0235	Printing Started	Record stage start/completion timestamps, operator, remarks and exceptions for printing started.
ALB-FR-0236	Printing Completed	Record stage start/completion timestamps, operator, remarks and exceptions for printing completed.
ALB-FR-0237	Lamination	Record stage start/completion timestamps, operator, remarks and exceptions for lamination.
ALB-FR-0238	Mounting	Record stage start/completion timestamps, operator, remarks and exceptions for mounting.
ALB-FR-0239	Binding	Record stage start/completion timestamps, operator, remarks and exceptions for binding.
ALB-FR-0240	Cover Production	Record stage start/completion timestamps, operator, remarks and exceptions for cover production.
ALB-FR-0241	Album Assembly	Record stage start/completion timestamps, operator, remarks and exceptions for album assembly.
ALB-FR-0242	Box Production	Record stage start/completion timestamps, operator, remarks and exceptions for box production.
ALB-FR-0243	Packaging	Record stage start/completion timestamps, operator, remarks and exceptions for packaging.
ALB-FR-0244	Production Completed	Record stage start/completion timestamps, operator, remarks and exceptions for production completed.
ALB-FR-0245	Sent to QC	Record stage start/completion timestamps, operator, remarks and exceptions for sent to qc.

12.4 Production Exceptions & Vendor Outsourcing

Requirement ID	Field / Action	Functional Behaviour
ALB-FR-0246	Material Shortage	Place job on hold, record material and expected availability, notify Admin.
ALB-FR-0247	Reprint Required	Create reprint event linked to reason, quantity, wastage and responsible stage.
ALB-FR-0248	Outsourced Vendor	Capture vendor, sent date, expected return, actual return and vendor cost.
ALB-FR-0249	Courier / AWB	Track external movement of job/material.
ALB-FR-0250	Production Notes	Maintain shift/operator notes.
ALB-FR-0251	Wastage Quantity	Record damaged/rejected sheets and reason for costing/reporting.

13. QC Department & Rework Management

13.1 QC Dashboard

Requirement ID	Widget / Queue	Type	Functional Behaviour
ALB-FR-0252	Awaiting QC	KPI	Production-complete jobs pending inspection.
ALB-FR-0253	QC In Progress	KPI	Active inspections.
ALB-FR-0254	Passed	KPI	QC-passed jobs for selected period.
ALB-FR-0255	Failed	KPI	Jobs failing QC.
ALB-FR-0256	Rework Pending	KPI	Jobs returned to production/design/grading.
ALB-FR-0257	Ready for Delivery	KPI	QC-passed and delivery-ready jobs.
ALB-FR-0258	QC SLA Breached	KPI	Overdue inspections.
ALB-FR-0259	Top Defects	Chart	Most frequent defect reasons.

13.2 QC Checklist

Requirement ID	Checklist Item	Result Type	Functional Behaviour
ALB-FR-0260	Colour Accuracy	Pass/Fail/N/A	Compare print colour to approved output/reference.
ALB-FR-0261	Exposure / Density	Pass/Fail/N/A	Check visible print consistency.
ALB-FR-0262	Print Sharpness	Pass/Fail/N/A	Check output quality and resolution.
ALB-FR-0263	Paper Type	Pass/Fail	Verify ordered paper stock.
ALB-FR-0264	Page Order	Pass/Fail	Verify sequence.
ALB-FR-0265	Cropping / Alignment	Pass/Fail	Check trims and positioning.
ALB-FR-0266	Lamination	Pass/Fail/N/A	Check bubbles, scratches and finish.
ALB-FR-0267	Binding	Pass/Fail	Check binding strength and lay-flat quality.
ALB-FR-0268	Cover	Pass/Fail	Verify material, print, personalization and finish.
ALB-FR-0269	Corners / Edges	Pass/Fail	Inspect physical damage and finishing.
ALB-FR-0270	Glue / Marks	Pass/Fail	Inspect manufacturing defects.
ALB-FR-0271	Album Box	Pass/Fail/N/A	Verify ordered box and personalization.
ALB-FR-0272	Quantity	Pass/Fail	Verify number of albums/copies.
ALB-FR-0273	Customer / Order Match	Pass/Fail	Prevent wrong-customer dispatch.

13.3 QC Failure & Rework

Requirement ID	Field / Action	Functional Behaviour
ALB-FR-0274	Fail QC	Requires defect code, severity, photo/evidence and rework routing.
ALB-FR-0275	Defect Department	Select Printing / Designing / Colour Grading / Other.
ALB-FR-0276	Rework Quantity	Record affected copies/pages where applicable.
ALB-FR-0277	Rework Note	Structured explanation sent to responsible department.
ALB-FR-0278	Evidence	Attach photo/video/document evidence.
ALB-FR-0279	Request Recheck	Creates linked second QC after rework completion.
ALB-FR-0280	Pass QC	Finalizes checklist and releases to delivery queue.
ALB-FR-0281	Manager Override	Elevated permission, reason and immutable audit required.

14. Delivery, Dispatch & Order Closure

14.1 Delivery Fields

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0282	Delivery Mode	Dropdown	M	Pickup / Courier / Company Delivery / Third-party.
ALB-FR-0283	Delivery Address	Textarea	C	Required for dispatch.
ALB-FR-0284	Courier Partner	Dropdown/text	C	Required for courier.
ALB-FR-0285	AWB / Tracking Number	Text	C	Required when available.
ALB-FR-0286	Dispatch Date	Datetime	C	Recorded at dispatch.
ALB-FR-0287	Expected Delivery	Date	O	Customer ETA.
ALB-FR-0288	Delivered Date	Datetime	C	Recorded on delivery confirmation.
ALB-FR-0289	Received By	Text	O	Recipient name.
ALB-FR-0290	Proof of Delivery	Attachment	O	Photo/signature/document.
ALB-FR-0291	Delivery Remarks	Textarea	O	Exception notes.
ALB-FR-0292	Delivery Status	Status	M	Ready / Dispatched / Delivered / Failed / Returned.

14.2 Closure Rules

Requirement ID	Business Rule
ALB-FR-0293	Order shall not be closed until all required production stages and QC are complete.
ALB-FR-0294	If policy requires full payment before delivery, dispatch shall be blocked while balance is outstanding unless Admin authorizes override.
ALB-FR-0295	A delivered order may remain financially open when company policy permits credit customers; operational and financial closure statuses must remain distinct.
ALB-FR-0296	Order closure shall preserve complete audit history, files, versions, approvals, invoices, payment history, QC evidence and delivery proof.

15. Accounts, Pricing, Payments & Invoicing

15.1 Order Financial Fields

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0297	Base Album Price	Currency	M	Base product amount.
ALB-FR-0298	Design Charges	Currency	C	Applied for design service.
ALB-FR-0299	Colour Grading Charges	Currency	O	Configured service charge.
ALB-FR-0300	Printing Charges	Currency	M	Printing production amount.
ALB-FR-0301	Cover Charges	Currency	O	Premium cover add-on.
ALB-FR-0302	Box Charges	Currency	O	Album box amount.
ALB-FR-0303	Courier Charges	Currency	O	Delivery charge.
ALB-FR-0304	Other Charges	Currency	O	Authorized additional amount.
ALB-FR-0305	Discount	Currency/%	O	Permission controlled.
ALB-FR-0306	Taxable Amount	Calculated	M	Amount before tax.
ALB-FR-0307	GST / Tax	Calculated	M	Tax breakup.
ALB-FR-0308	Grand Total	Calculated	M	Final payable amount.
ALB-FR-0309	Advance Received	Calculated	M	Sum of qualifying receipts.
ALB-FR-0310	Balance Due	Calculated	M	Grand total minus net payments/refunds.

15.2 Payments

Requirement ID	Field / Column	Type	M/O	Functional Behaviour
ALB-FR-0311	Payment ID	Auto ID	M	Unique receipt transaction.
ALB-FR-0312	Order / Customer	Link	M	Links payment.
ALB-FR-0313	Amount	Currency	M	Positive amount.
ALB-FR-0314	Payment Date	Datetime	M	Server/user date with permission.
ALB-FR-0315	Method	Dropdown	M	Cash/UPI/Bank/Card/Cheque/Other.
ALB-FR-0316	Reference Number	Text	C	Transaction/cheque reference.
ALB-FR-0317	Received By	User	M	Employee receiving payment.
ALB-FR-0318	Receipt Number	Auto	M	Printable customer receipt.
ALB-FR-0319	Status	Status	M	Pending/Cleared/Failed/Refunded.
ALB-FR-0320	Attachment	File	O	Payment proof.

15.3 Invoicing & Accounts Actions

Requirement ID	Action / Document	Functional Behaviour
ALB-FR-0321	Quotation	Generate commercial proposal before order confirmation.
ALB-FR-0322	Proforma Invoice	Generate provisional invoice.
ALB-FR-0323	Tax Invoice	Generate GST-compliant invoice using configured numbering.
ALB-FR-0324	Payment Receipt	Generate receipt for successful payment.
ALB-FR-0325	Credit Note	Create authorized reduction/refund adjustment.
ALB-FR-0326	Record Refund	Capture refund amount, method, reason and approval.
ALB-FR-0327	Outstanding Report	Show customer/order-wise unpaid balances and ageing.
ALB-FR-0328	Expense Entry	Record approved operational expense and category.
ALB-FR-0329	Lock Financial Document	Prevent silent edits after issue; corrections use amendment/credit note.

16. Admin Control Center

16.1 Admin Dashboard KPIs

Requirement ID	KPI / Widget	Functional Behaviour
ALB-FR-0330	Orders Received	Today / selected period.
ALB-FR-0331	In Production	All active production orders.
ALB-FR-0332	Colour Grading Pending	Open grading jobs.
ALB-FR-0333	Designing Pending	Open design jobs.
ALB-FR-0334	Client Approval Pending	Proofs awaiting client response.
ALB-FR-0335	Corrections Pending	Open admin/client corrections.
ALB-FR-0336	Printing Pending	Jobs queued/active in printing.
ALB-FR-0337	QC Pending	Jobs awaiting inspection.
ALB-FR-0338	Ready for Delivery	QC-passed orders.
ALB-FR-0339	Delivered Today	Completed deliveries today.
ALB-FR-0340	Payments Received	Collections for selected period.
ALB-FR-0341	Pending Dues	Outstanding receivables.
ALB-FR-0342	Overdue Orders	Promised delivery date breached.
ALB-FR-0343	Department SLA Risk	Jobs at risk by department.

16.2 Administrative Controls

Requirement ID	Action / Control	Functional Behaviour
ALB-FR-0344	Create / Disable User	Manage employee access lifecycle.
ALB-FR-0345	Assign Role	Map role and fine-grained permissions.
ALB-FR-0346	Assign Department	Controls work queues and visibility.
ALB-FR-0347	Reassign Job	Move work between eligible staff with reason.
ALB-FR-0348	Override Workflow	Skip/reopen stage only with elevated permission and audit.
ALB-FR-0349	Reopen Closed Order	Requires reason and records amendment event.
ALB-FR-0350	Manage Masters	Create/disable configurable options without deleting historical references.
ALB-FR-0351	Manage SLA	Set target hours/days by department, priority and order type.
ALB-FR-0352	Manage Templates	Configure WhatsApp/email/SMS templates.
ALB-FR-0353	Manage Branches	Configure office/production locations if multi-branch.
ALB-FR-0354	View Audit Log	Search all critical events.

Requirement ID	Action / Control	Functional Behaviour
ALB-FR-0355	Export Reports	Generate permission-controlled exports.

17. Notifications, WhatsApp, Email & Communication Logs

17.1 Triggered Notifications

Requirement ID	Trigger	Recipient	Functional Behaviour
ALB-FR-0356	Order Created	Customer / Admin	Send order acknowledgement with Order ID.
ALB-FR-0357	Colour Grading Assigned	Colour Grader	New job notification.
ALB-FR-0358	Grading Submitted	Admin	Approval required.
ALB-FR-0359	Design Assigned	Designer	New design task.
ALB-FR-0360	Design Submitted	Admin	Review required.
ALB-FR-0361	Proof Ready	Customer	Send secure review link.
ALB-FR-0362	Correction Received	Admin / Designer	Notify correction task.
ALB-FR-0363	Final Approval	Admin / Printing	Release notification.
ALB-FR-0364	Printing Completed	QC	QC job notification.
ALB-FR-0365	QC Passed	Reception / Admin	Order ready for delivery.
ALB-FR-0366	Payment Due	Customer / Accounts	Reminder based on configured schedule.
ALB-FR-0367	Dispatched	Customer	Courier/tracking communication.
ALB-FR-0368	Delivered	Customer / Admin	Delivery confirmation.

17.2 Communication Log

Requirement ID	Field	Functional Behaviour
ALB-FR-0369	Channel	WhatsApp / Email / SMS / Call / Internal Note.
ALB-FR-0370	Direction	Inbound / Outbound.
ALB-FR-0371	Template	Stores template used where applicable.
ALB-FR-0372	Recipient	Masked according to permission.
ALB-FR-0373	Sent At	Timestamp.
ALB-FR-0374	Delivery Status	Queued/Sent/Delivered/Read/Failed where provider supports.
ALB-FR-0375	Linked Order	Associates communication to order.
ALB-FR-0376	Message Summary	Stores safe operational summary; full content per retention policy.
ALB-FR-0377	Attachment	Stores permitted documents sent/received.

18. Production Pipeline, SLA, Priority & Escalation

18.1 Visual Pipeline

Admin should have a Kanban-style production pipeline where each card shows Order ID, customer, event, due date, priority, current owner, current stage, elapsed time and warning badges. Default columns: New Order -> Files Received -> Colour Grading -> Admin Approval -> Designing -> Admin Review -> Client Review -> Correction -> Final Approved -> Printing -> QC -> Ready for Delivery -> Delivered.

Requirement ID	Pipeline Capability	Functional Behaviour
ALB-FR-0378	Stage Counts	Display live order counts per pipeline stage.
ALB-FR-0379	Card Drill-down	Open order detail without losing filter context.
ALB-FR-0380	Permission-based Drag	Allow stage movement only where workflow and permission permit.
ALB-FR-0381	Blocked Badge	Show payment/file/approval/material/QC blockers.
ALB-FR-0382	Ageing	Show time spent in current stage.
ALB-FR-0383	Due Risk	Colour/badge indication for On Time / Due Today / Delayed / Critical.
ALB-FR-0384	Owner	Show assigned department user.
ALB-FR-0385	Quick Reassign	Admin can reassign with reason.
ALB-FR-0386	Saved Views	Filter by branch, order type, priority, customer, employee and date.

18.2 SLA Rules

Requirement ID	SLA Rule	Functional Behaviour
ALB-FR-0387	Colour Grading TAT	Configurable target from assignment/start to submission/approval.
ALB-FR-0388	Design TAT	Configurable target from assignment to draft submission.
ALB-FR-0389	Admin Review TAT	Configurable target for grading/design approval.
ALB-FR-0390	Client Review TAT	Tracked separately and may pause internal SLA if configured.
ALB-FR-0391	Printing TAT	Target from print release to production complete.
ALB-FR-0392	QC TAT	Target from QC receipt to pass/fail decision.
ALB-FR-0393	Delivery TAT	Target from Ready for Delivery to dispatch/hand-over.
ALB-FR-0394	Escalation Threshold	Warn at configurable percentage of SLA consumption.
ALB-FR-0395	SLA Pause	Only allowed for configured reasons such as waiting client/material/payment.
ALB-FR-0396	SLA Breach	Creates escalation/notification and appears in reports.

19. Reports, Analytics & Operational Dashboards

19.1 Operational Reports

Requirement ID	Report	Functional Behaviour
ALB-FR-0397	Order Register	All orders with type, customer, date, current stage, owner, due date and status.
ALB-FR-0398	Department Pending	Open work by Colour Grading, Designing, Printing, QC.
ALB-FR-0399	Employee Productivity	Jobs completed, average TAT, rework count and SLA performance.
ALB-FR-0400	Client Approval TAT	Time from proof sent to final approval.
ALB-FR-0401	Correction Report	Corrections per order/revision/page/designer and resolution time.
ALB-FR-0402	Printing Production	Throughput by stage, operator, product and date.
ALB-FR-0403	Wastage / Reprint	Material loss, reprints, reason and cost impact.
ALB-FR-0404	QC Defect	Pass/fail rate, defects, severity, responsible stage and rework.
ALB-FR-0405	Delivery	Ready, dispatched, delivered, failed and ageing.
ALB-FR-0406	SLA Breach	Breached orders by department, owner and priority.

19.2 Financial Reports

Requirement ID	Report	Functional Behaviour
ALB-FR-0407	Revenue	Invoice/order revenue by day/month/customer/product.
ALB-FR-0408	Collections	Payments received by method/date/customer.
ALB-FR-0409	Outstanding	Balance due with ageing buckets.
ALB-FR-0410	Customer Ledger	Invoices, payments, credits, balance.
ALB-FR-0411	Service Revenue	Design, grading, printing, cover, box and delivery revenue split.
ALB-FR-0412	Vendor Cost	Outsourced printing/production cost.
ALB-FR-0413	Expense Summary	Operational expenses by category.
ALB-FR-0414	Profitability	Revenue less configured direct production/vendor costs.
ALB-FR-0415	Tax Report	GST/tax summary for configured period.

19.3 Export & Scheduling

Requirement ID	Action	Functional Behaviour
ALB-FR-0416	Export CSV/XLSX/PDF	Permission-controlled export using current filters.
ALB-FR-0417	Schedule Report	Admin can schedule periodic report to permitted internal

Requirement ID	Action	Functional Behaviour
		recipients.
ALB-FR-0418	Drill-down	KPI/chart click opens supporting records.
ALB-FR-0419	Saved Report View	Persist filters/columns for user/team.
ALB-FR-0420	Audit Export	Exports are logged with user, filters, format and expiry.

20. Search, Audit, Security & Data Governance

20.1 Global Search

Requirement ID	Search Key	Functional Behaviour
ALB-FR-0421	Order ID	Exact/partial match.
ALB-FR-0422	Customer / Studio Name	Permission-scoped text search.
ALB-FR-0423	Mobile / WhatsApp	Masked results if user lacks full view permission.
ALB-FR-0424	Bride / Groom	Search wedding context.
ALB-FR-0425	Event Name	Search project label.
ALB-FR-0426	Invoice / Receipt Number	Financial lookup for permitted users.
ALB-FR-0427	Tracking Number	Delivery lookup.
ALB-FR-0428	File Name	Search order files subject to access.

20.2 Audit Events

Requirement ID	Event	Functional Behaviour
ALB-FR-0429	Login / Logout	Record authentication event.
ALB-FR-0430	Order Create / Edit	Store changed fields and values.
ALB-FR-0431	Assignment	Store from/to user and reason.
ALB-FR-0432	Approval / Rejection	Store decision, role, timestamp and note.
ALB-FR-0433	File Upload / Version / Lock	Store file lifecycle.
ALB-FR-0434	Client Approval	Store version and approval metadata.
ALB-FR-0435	Payment / Refund / Invoice	Financial event audit.
ALB-FR-0436	QC Decision	Store checklist result, evidence and disposition.
ALB-FR-0437	Delivery	Store dispatch/delivery proof event.
ALB-FR-0438	Permission Change	Store old/new permissions and admin actor.
ALB-FR-0439	Override	Requires reason and elevated audit detail.

20.3 Security & Data Rules

Requirement ID	Business Rule
ALB-FR-0440	Role-based access shall be enforced at both UI and API/service layers.
ALB-FR-0441	Sensitive financial and customer fields shall support field-level permission and masking.
ALB-FR-0442	Final approved files and finalized financial documents shall not be silently overwritten.

Requirement ID	Business Rule
ALB-FR-0443	Downloads and exports may be watermarked/logged according to policy.
ALB-FR-0444	Backups, retention and deletion shall follow configured company policy and legal requirements.
ALB-FR-0445	All timestamps shall be stored consistently and rendered in configured business timezone.
ALB-FR-0446	System shall support secure password storage, HTTPS, CSRF protection, input validation and secure upload handling.
ALB-FR-0447	Critical actions may require re-authentication/MFA and reason capture.

21. Role Permission Matrix

Requirement ID	Module / Action	Admin	Reception	Colour Grading	Designing	Printing	QC	Accounts
ALB-FR-0448	Customer Create/Edit	Full	Create/Edit	View	View	View	View	View
ALB-FR-0449	Order Create/Edit	Full	Create/Edit	View	View	View	View	View
ALB-FR-0450	Printing Requirement	Full	Create/Edit	View	View	View	View	View
ALB-FR-0451	Colour Grading Work	Full	View	Work	View	View	View	No
ALB-FR-0452	Approve Colour Grading	Approve	No	No	No	No	No	No
ALB-FR-0453	Album Design Work	Full	View	View	Work	View	View	No
ALB-FR-0454	Approve Design	Approve	No	No	No	No	No	No
ALB-FR-0455	Send Client Proof	Full	Limited	No	No	No	No	No
ALB-FR-0456	Client Correction View	Full	View	No	Work	No	No	No
ALB-FR-0457	Release to Printing	Approve	No	No	No	No	No	No
ALB-FR-0458	Printing Production	Full	View	No	View	Work	View	No
ALB-FR-0459	QC Inspection	Full	View	No	View	View	Work	No
ALB-FR-0460	Delivery	Full	Update	No	No	Update	Update	View
ALB-FR-0461	Payments	Full	Receive limited	No	No	No	No	Full
ALB-FR-0462	Invoices / Refunds	Full	View/receipt	No	No	No	No	Full
ALB-FR-0463	Reports	Full	Limited	Dept	Dept	Dept	Dept	Finance
ALB-FR-0464	User / Role Admin	Full	No	No	No	No	No	No

22. End-to-End Workflow Definitions

22.1 Design + Album Printing Workflow

Requirement ID	Step	Owner	Workflow Behaviour	Resulting Status
ALB-FR-0465	1	Reception	Create customer/order; capture printing requirement; receive source files.	Files Received
ALB-FR-0466	2	Colour Grading	Process selected photos and submit graded output.	Pending Admin Grading Approval
ALB-FR-0467	3	Admin	Approve grading or return correction.	Ready for Designing
ALB-FR-0468	4	Designing	Create album draft and submit.	Pending Admin Design Review
ALB-FR-0469	5	Admin	Approve proof or request correction.	Ready for Client Review
ALB-FR-0470	6	Client	Review pages, approve or submit corrections.	Correction / Final Approval
ALB-FR-0471	7	Designing + Admin	Resolve corrections and republish revision as needed.	Client Review
ALB-FR-0472	8	Client	Final approve album for printing.	Print Locked
ALB-FR-0473	9	Admin	Release locked approved version.	Waiting for Printing
ALB-FR-0474	10	Printing	Execute production stages and packaging.	Sent to QC
ALB-FR-0475	11	QC	Inspect, pass or return rework.	Ready for Delivery
ALB-FR-0476	12	Reception / Delivery	Dispatch or hand over album.	Delivered
ALB-FR-0477	13	Accounts / Admin	Confirm financial/operational closure rules.	Closed

22.2 Album Printing Only Workflow

Requirement ID	Step	Owner	Workflow Behaviour	Resulting Status
ALB-FR-0478	1	Reception	Create order and capture exact printing requirement.	Files Pending / Received
ALB-FR-0479	2	Reception/Admin	Verify print-ready final file.	Ready for Printing
ALB-FR-0480	3	Printing	Print, finish, bind, assemble and package.	Sent to QC
ALB-FR-0481	4	QC	Inspect album against requirement and approved file.	Ready for Delivery / Rework
ALB-FR-0482	5	Reception / Delivery	Hand over or dispatch.	Delivered
ALB-FR-0483	6	Accounts / Admin	Close after required payment and operational checks.	Closed

23. Status Model & State Transitions

Requirement ID	Status Domain	Allowed Values	Functional Behaviour
ALB-FR-0484	Main Order	New / Active Production / Awaiting Client / Printing / QC / Ready / Delivered / Closed / Cancelled	High-level lifecycle.
ALB-FR-0485	File Intake	Pending / Partial / Complete / On Hold	Tracks source-file readiness.
ALB-FR-0486	Colour Grading	Not Required / Pending / In Progress / Admin Review / Revision / Approved	Creative pre-processing state.
ALB-FR-0487	Design	Not Required / Pending / Designing / Admin Review / Client Review / Correction / Approved	Album design state.
ALB-FR-0488	Printing	Not Required / Waiting / File Prep / Printing / Finishing / Assembly / Packaging / Completed / Sent to QC	Production state.
ALB-FR-0489	QC	Pending / In Progress / Failed / Rework / Recheck / Passed	Inspection state.
ALB-FR-0490	Delivery	Not Ready / Ready / Dispatched / Delivered / Failed / Returned	Fulfilment state.
ALB-FR-0491	Payment	Unpaid / Partially Paid / Paid / Overpaid / Refunded / Credit	Financial state.
ALB-FR-0492	Order Closure	Open / Operationally Complete / Financially Complete / Closed	Separates operation and finance closure.

23.2 Transition Rules

Requirement ID	Business Rule
ALB-FR-0493	Only valid configured transitions shall be accepted; API calls must enforce the same rules as UI.
ALB-FR-0494	Skipping a required stage requires elevated permission, reason and audit event.
ALB-FR-0495	Final client approval is mandatory before normal release of design-included orders to printing.
ALB-FR-0496	QC Passed is mandatory before Ready for Delivery unless Admin uses explicit override policy.
ALB-FR-0497	Closed orders are read-only for normal users; reopen requires elevated permission.
ALB-FR-0498	Cancellation shall preserve all prior work, payments, files and audit history and record cancellation reason.

24. Non-Functional Requirements

Requirement ID	Non-Functional Area	Requirement
ALB-FR-0499	Performance	Primary dashboard/list pages should load within acceptable operational target under expected concurrent usage; heavy reports may run as background jobs.
ALB-FR-0500	Availability	System should support business-hours production use with monitored uptime and recoverable failures.
ALB-FR-0501	Scalability	Architecture should support increasing customers, orders, files, branches and employees without data model redesign.
ALB-FR-0502	Security	HTTPS, secure auth, MFA support, least privilege, encryption at rest where applicable, secure upload validation and audit trails.
ALB-FR-0503	Reliability	Workflow transitions and financial postings must be atomic or recoverable; duplicate submissions should be idempotent where required.
ALB-FR-0504	Backup & Recovery	Configurable automated backups and tested restore process for database and critical file metadata.
ALB-FR-0505	Usability	Role-specific dashboards, consistent controls, clear status labels, responsive desktop/tablet layouts and accessible forms.
ALB-FR-0506	Browser Support	Current major Chrome/Edge/Safari versions as defined by product policy.
ALB-FR-0507	File Handling	Large file operations should support resumable/background upload where feasible and clear progress/failure states.
ALB-FR-0508	Observability	Application errors, background jobs, integration failures and critical workflow events should be logged and monitorable.
ALB-FR-0509	Localization	Currency INR and configurable date/time formats; architecture should not hard-code a single locale.
ALB-FR-0510	Data Retention	Configurable retention for operational, financial, audit and uploaded file classes.

25. Acceptance Criteria & UAT Traceability

25.1 Core UAT Scenarios

Requirement ID	UAT Scenario	Acceptance Criteria
ALB-FR-0511	Create Design + Printing Order	Reception can create customer/order, capture printing requirement and upload source files; workflow routes to grading.
ALB-FR-0512	Colour Grading Approval Loop	Grader submits, Admin rejects with correction, grader resubmits, Admin approves; history remains intact.
ALB-FR-0513	Design Approval Loop	Designer submits proof, Admin requests correction, designer resubmits and Admin publishes client proof.
ALB-FR-0514	Client Correction Loop	Client adds page corrections, designer resolves them, new revision is published and previous revision remains viewable.
ALB-FR-0515	Final Approval Lock	Client final approval locks revision; Printing sees only locked final file.
ALB-FR-0516	Printing Production	Printing records all applicable stages, operator, timestamps, notes and completes handoff to QC.
ALB-FR-0517	QC Rework	QC fails album with evidence, creates rework, production resolves, QC rechecks and passes.
ALB-FR-0518	Printing Only Order	Order bypasses grading/design while still preserving Not Required statuses and file verification.
ALB-FR-0519	Payment & Delivery Block	Configured unpaid-balance rule prevents delivery until payment or authorized override.
ALB-FR-0520	Delivery Completion	Courier/pickup data and proof of delivery are captured and status changes to Delivered.
ALB-FR-0521	Role Access	Each role sees only permitted modules/actions; denied actions return clear permission message.
ALB-FR-0522	Audit Traceability	Critical actions can be reconstructed by user, timestamp, old/new state and reason.
ALB-FR-0523	Report Reconciliation	Operational and financial reports reconcile to underlying orders/payments/statuses for sample period.

25.2 Definition of Done

Requirement ID	Business Rule
ALB-FR-0524	Every functional requirement shall map to at least one development story and one test case before production release.
ALB-FR-0525	All role permissions shall be validated with positive and negative UAT.
ALB-FR-0526	Critical workflows shall be tested for happy path, rejection/correction path, cancellation, retry and unauthorized access.
ALB-FR-0527	All issued reports shall be reconciled against source data for representative samples.
ALB-FR-0528	No unresolved critical security, data-loss, financial posting or workflow-bypass defect may remain at go-live.

Appendix A - Suggested Main Navigation

Menu	Purpose
Dashboard	Role-specific KPIs and queues
Orders	All permitted orders and search
Customers	Customer master and order history
Production Pipeline	Kanban across all lifecycle stages
Colour Grading	Department queue and jobs
Album Designing	Designer queue, revisions and corrections
Client Approvals	Proofs, feedback and final approvals
Printing	Production queue and stage tracking
Quality Control	QC queue, checklist, rework
Delivery	Ready/dispached/delivered orders
Payments	Receipts and dues
Invoices	Quotation/proforma/tax invoice/credit note
Expenses	Operational expense entries
Products & Pricing	Album products, materials, rates
Masters	Configurable options
Team & Roles	Users, departments, permissions
Reports	Operational and financial analytics
Notifications	Templates and delivery logs
Audit Logs	Immutable trace history
Settings	SLA, branch, integration and system configuration

Appendix B - Recommended Dashboard Stage Strip

New Order -> Files Received -> Colour Grading -> Admin Approval -> Designing -> Admin Review -> Client Review -> Correction -> Final Approved -> Printing -> QC -> Ready for Delivery -> Delivered

Total traceable functional requirements in this document: 528.